PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

Captura de Conformação de Cromossomo

Amanda Souza Câmara

Leibniz Institute of Plant Genetics and Crop Plant Research



26 de outubro de 2026

In Silico Genebank Proteomics

IPK LEIBNIZ-INSTITUT

Science

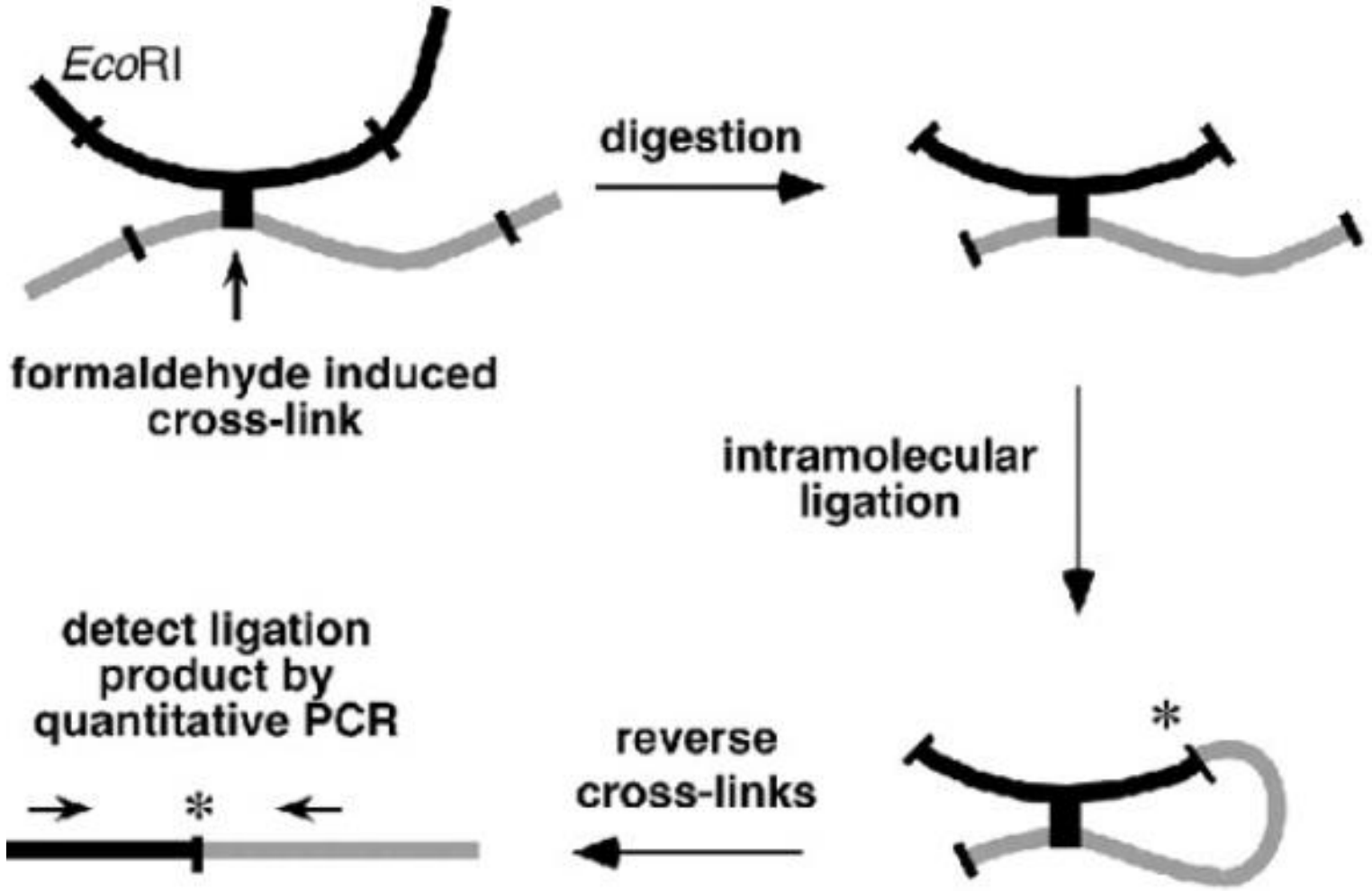
REPORTS

Capturing Chromosome Conformation

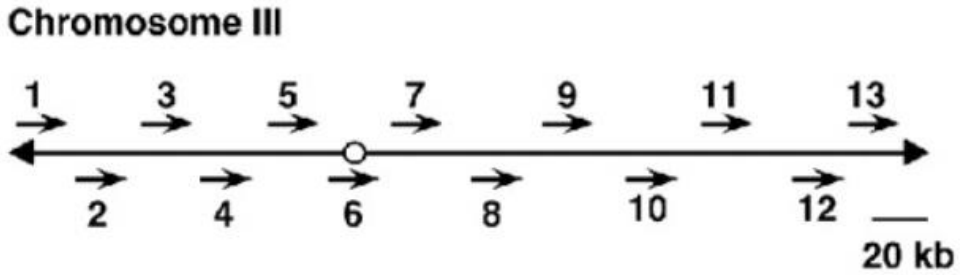
JOB DEKKER, KARSTEN RIPPE, MARTIJN DEKKER, AND NANCY KLECKNER Authors Info & Affiliations

SCIENCE • 15 Feb 2002 • Vol 295, Issue 5558 • pp. 1306-1311 • DOI: 10.1126/science.1067799

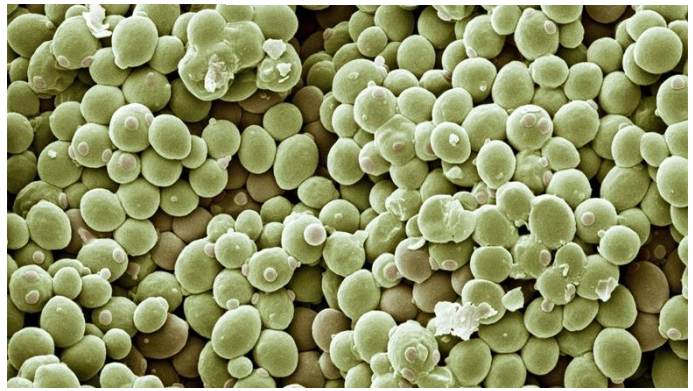
O experiment de 3C



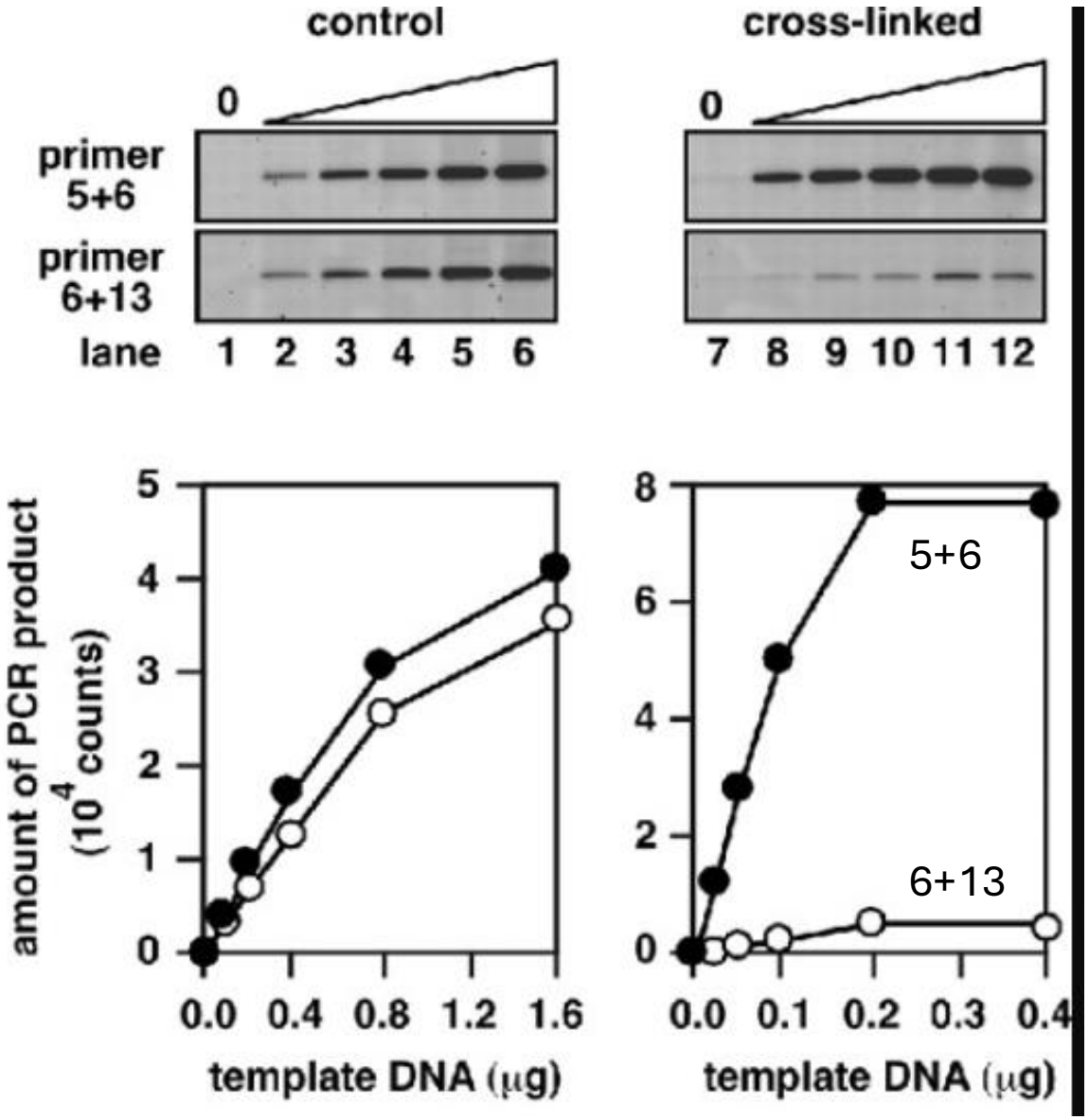
O experiment de 3C



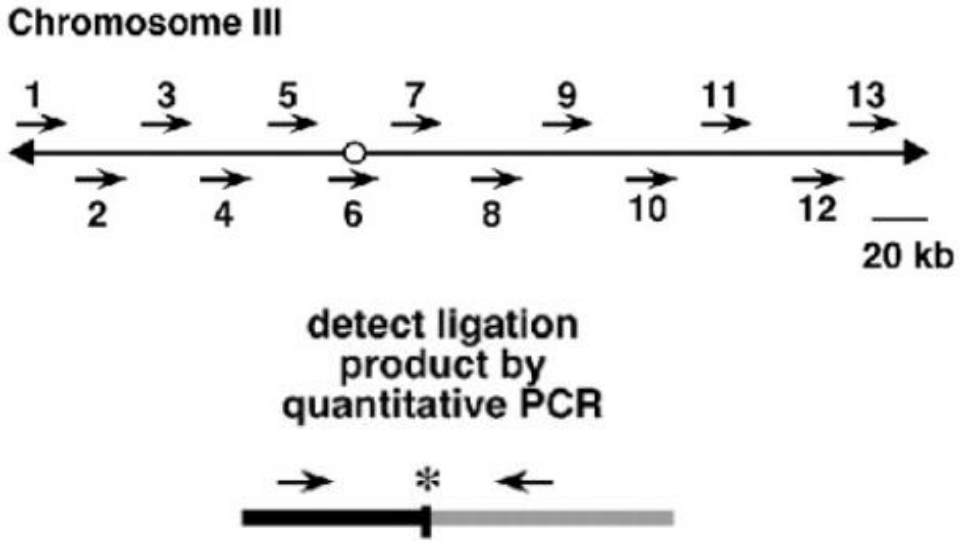


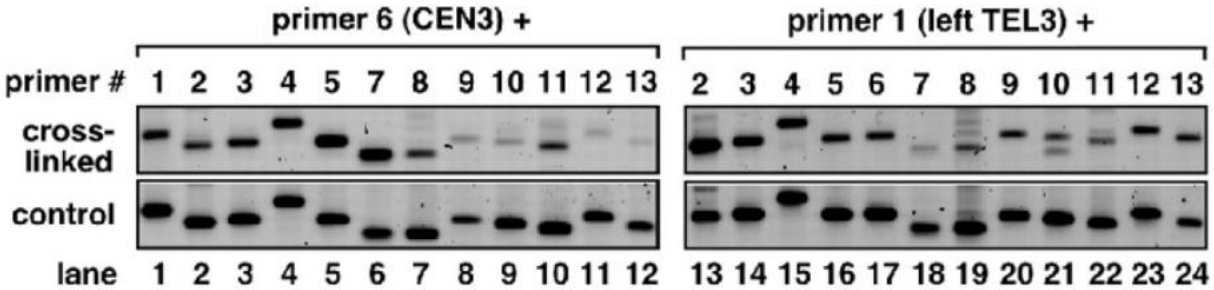


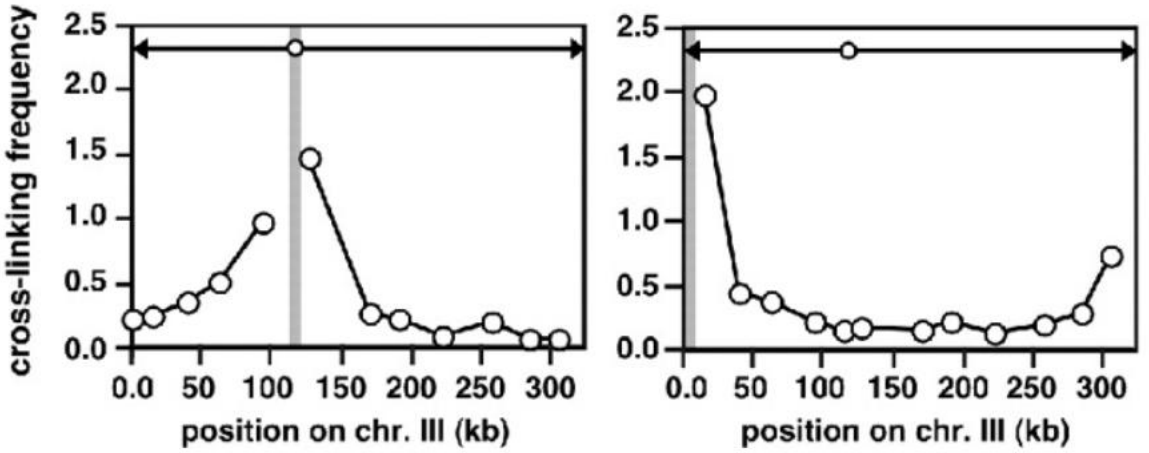
levedura

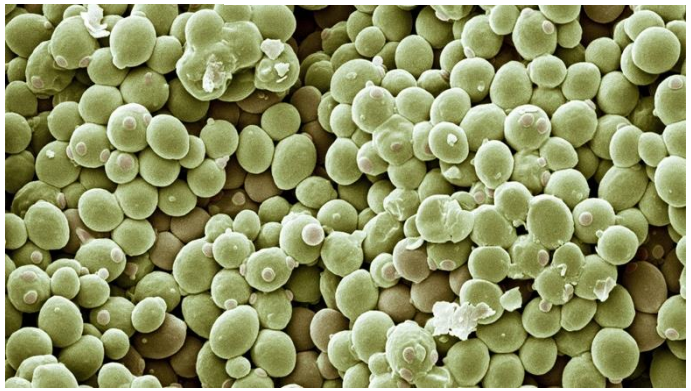


Quantificação pra todas as possíveis combinações de primers



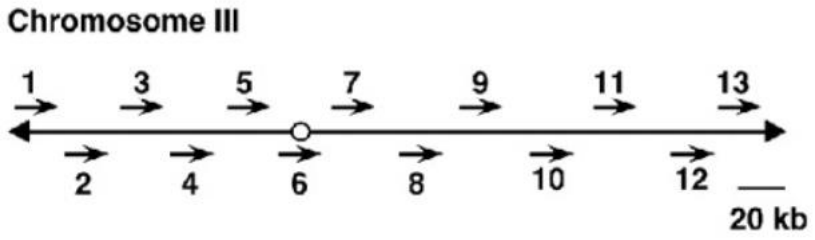




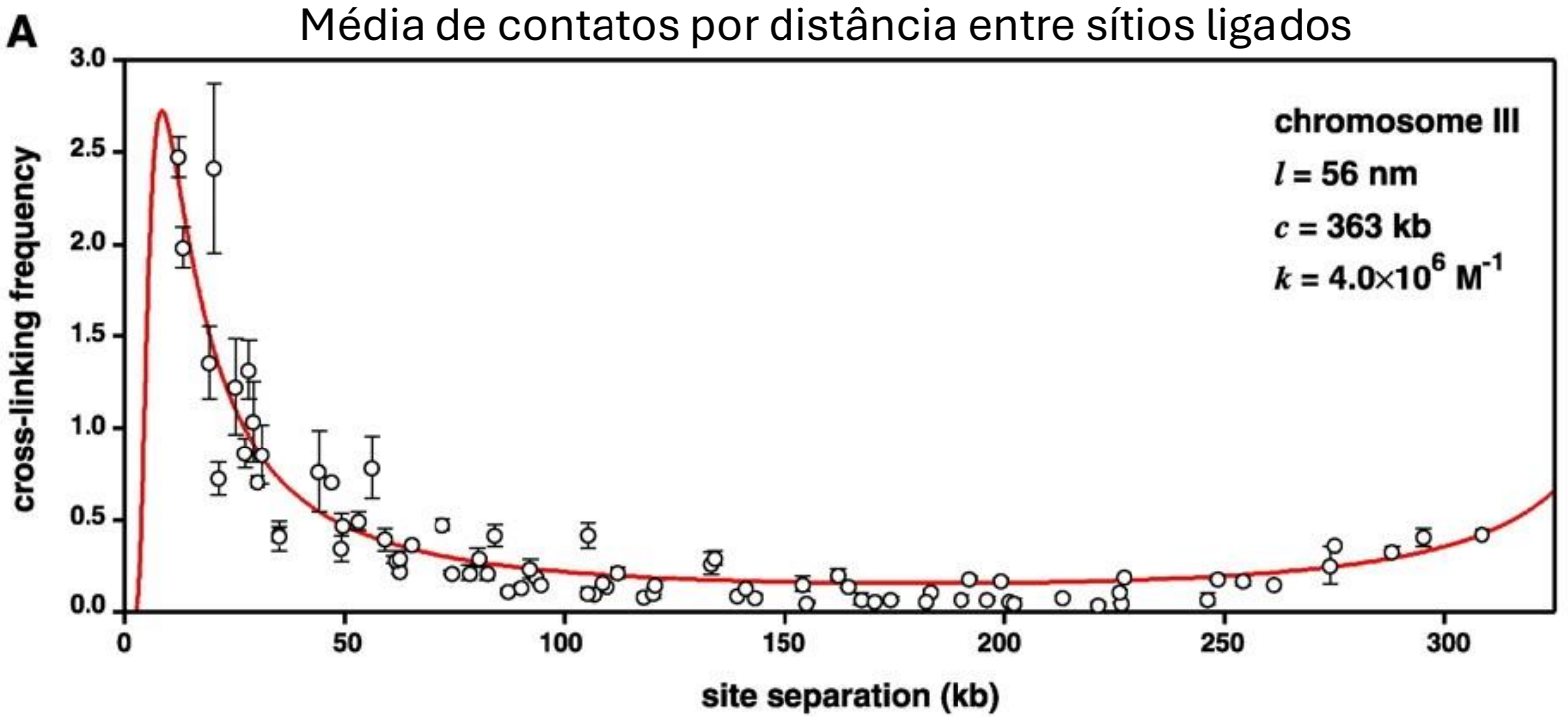


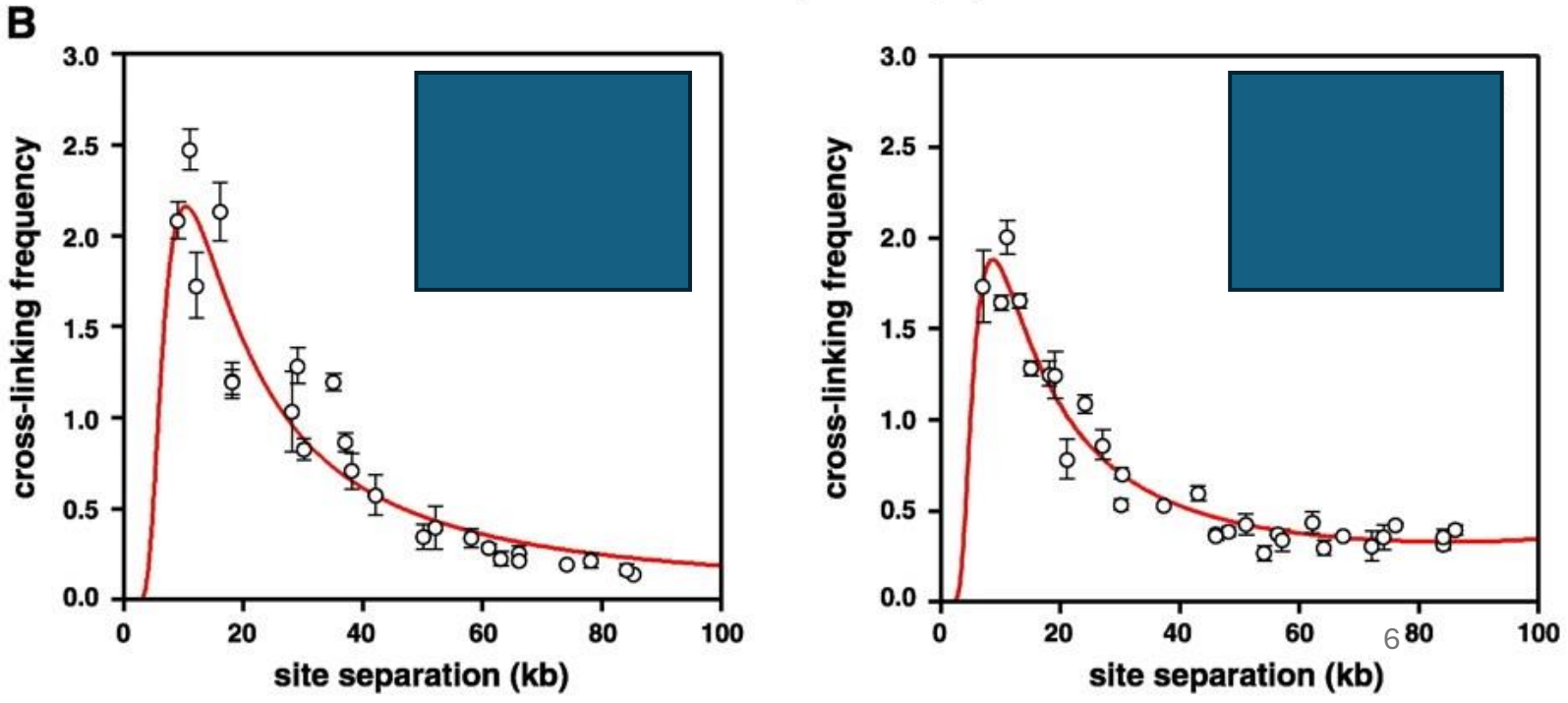
levedura

Frequência de contato

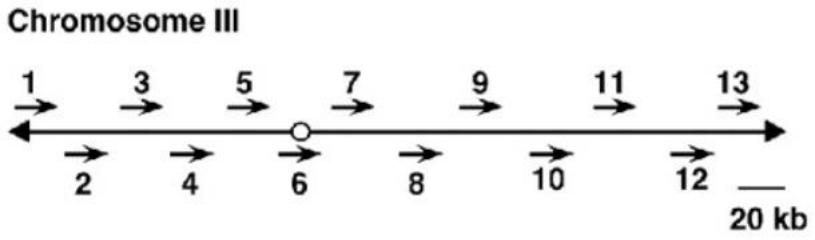


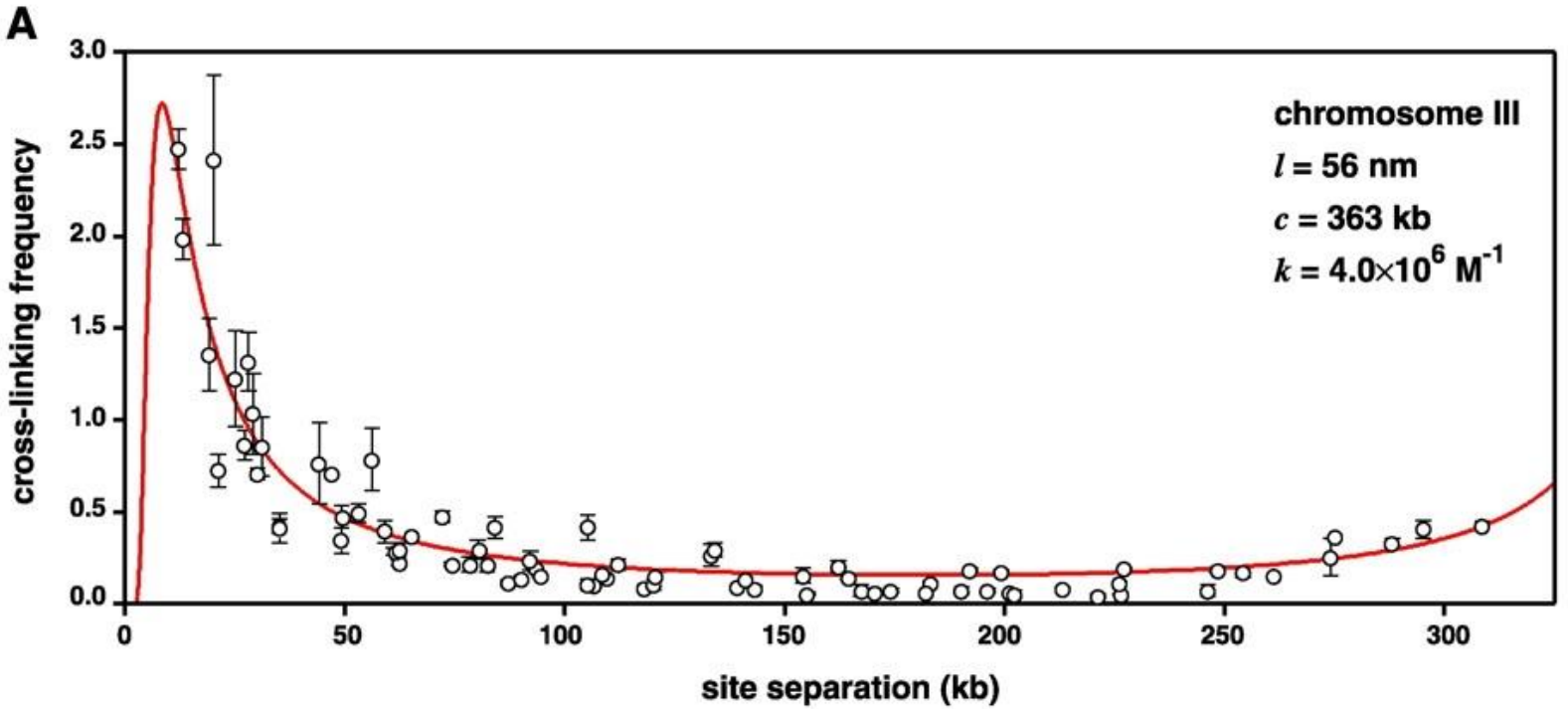
Média de contatos por distância entre sítios ligados

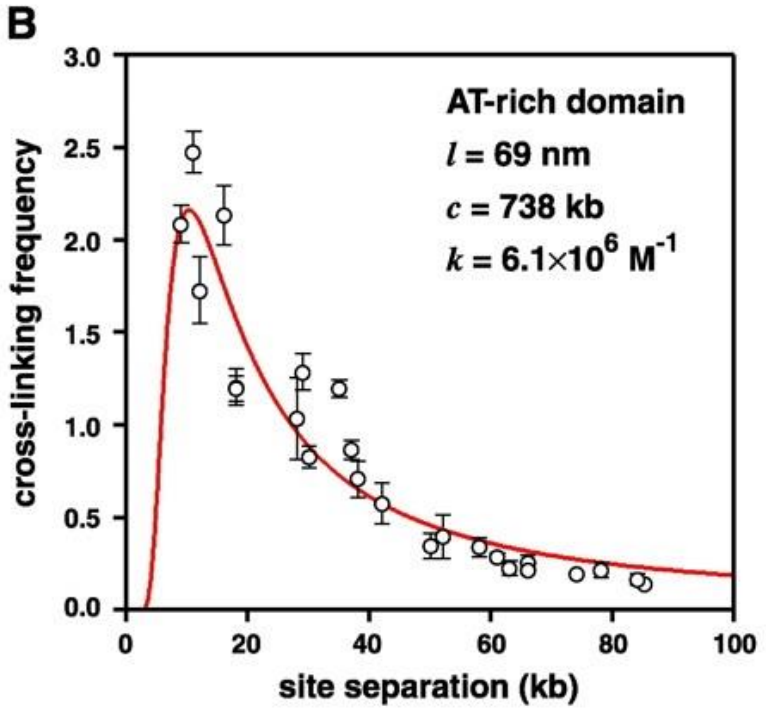


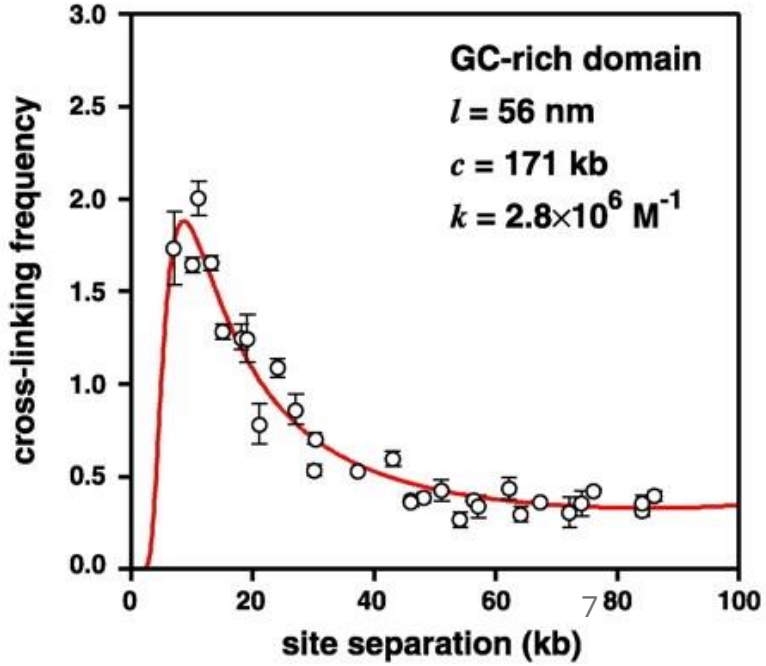


Frequência de contato



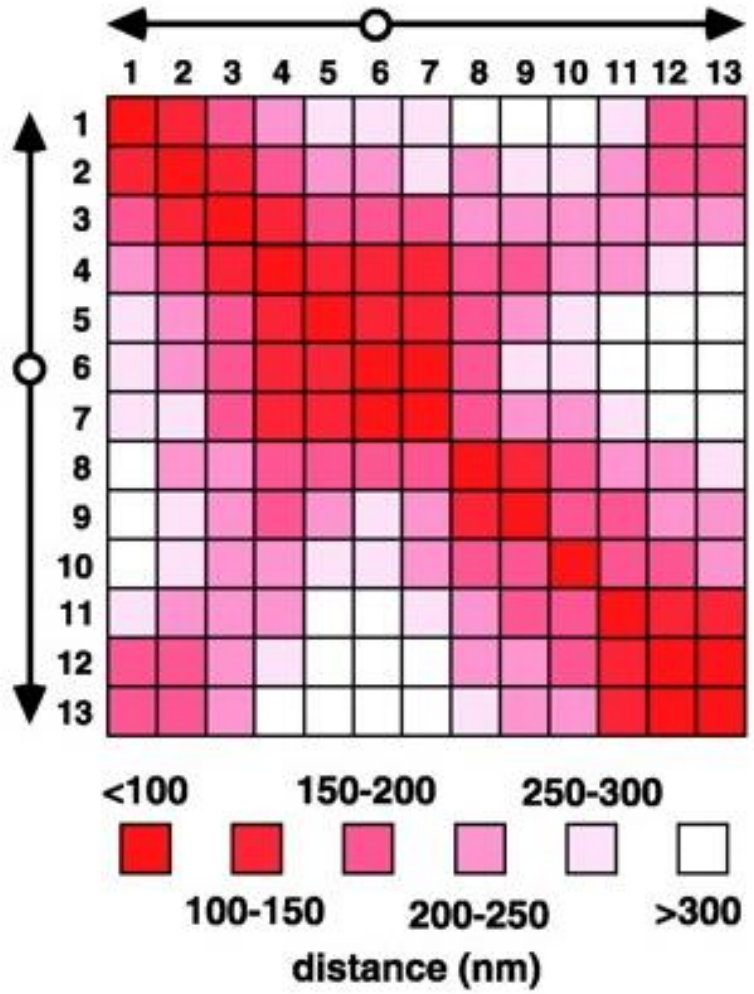






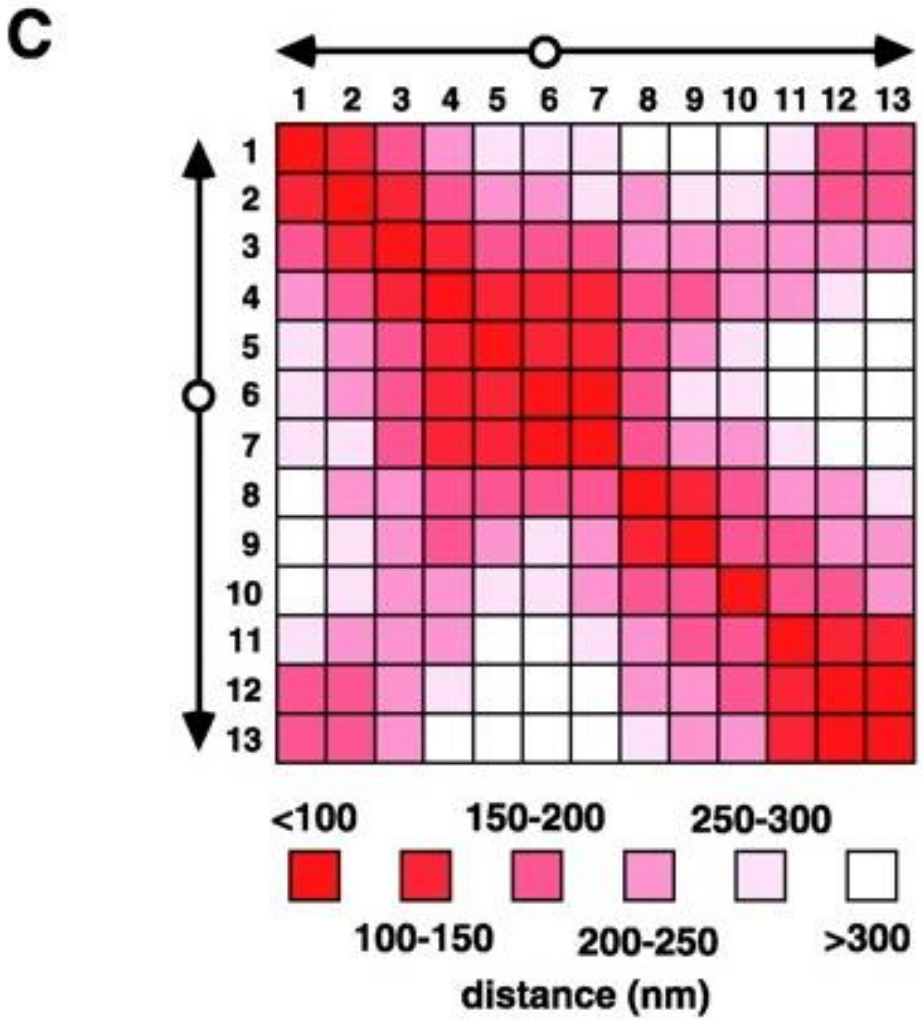
Mapa de contatos

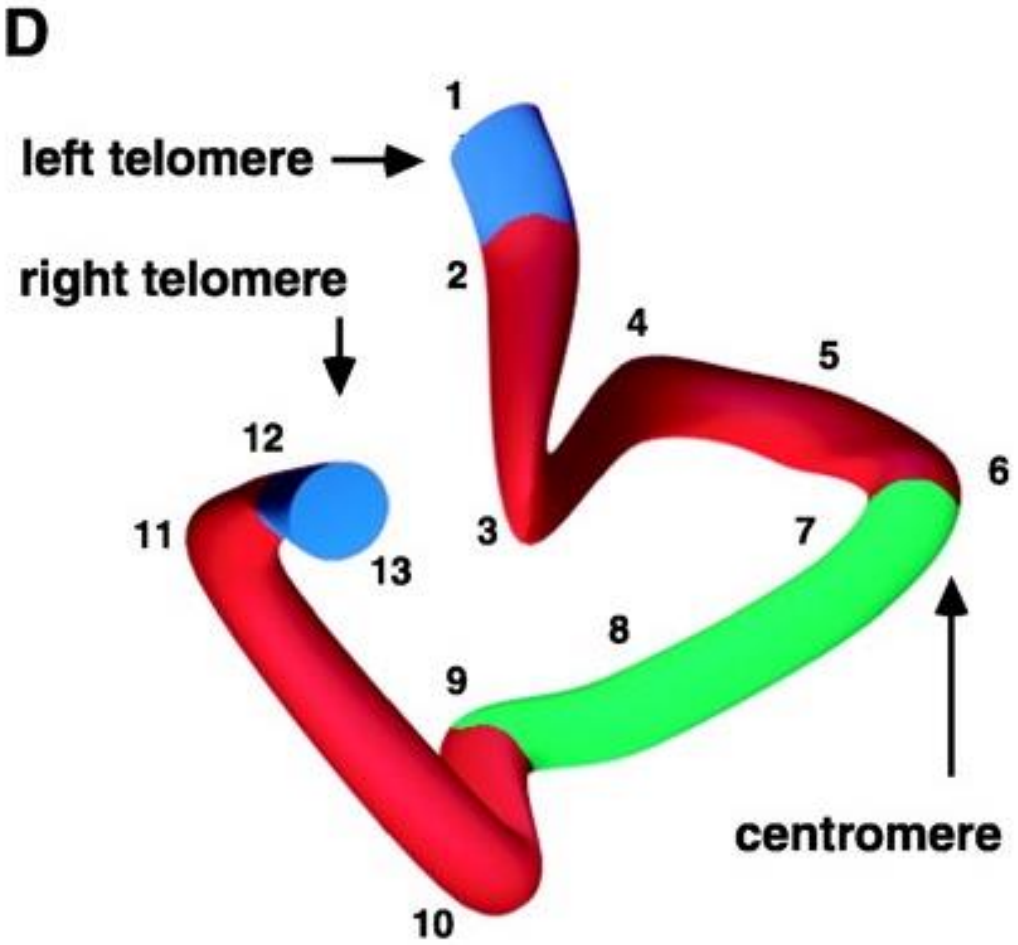
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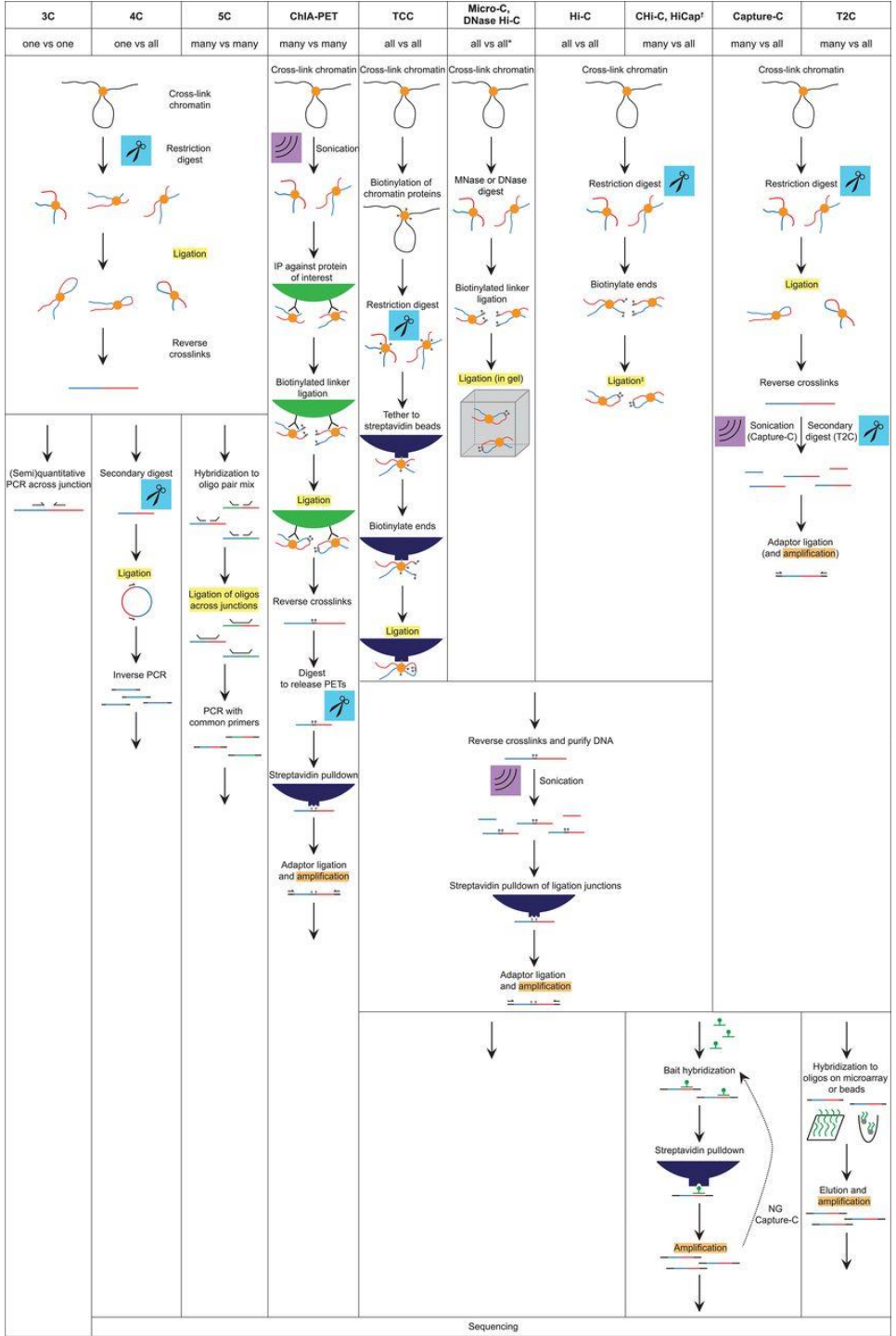


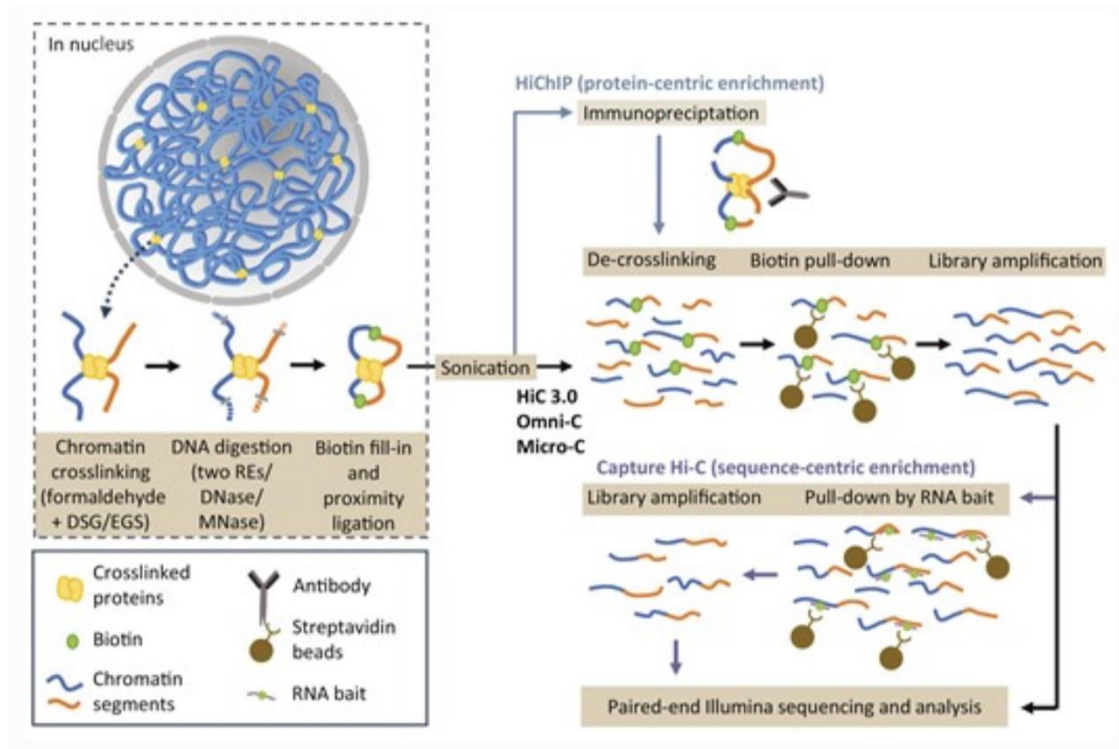
Mapa de contatos





Família 3C





Simková et al, 2024

Denker and de Laat, 2016

Science

REPORT

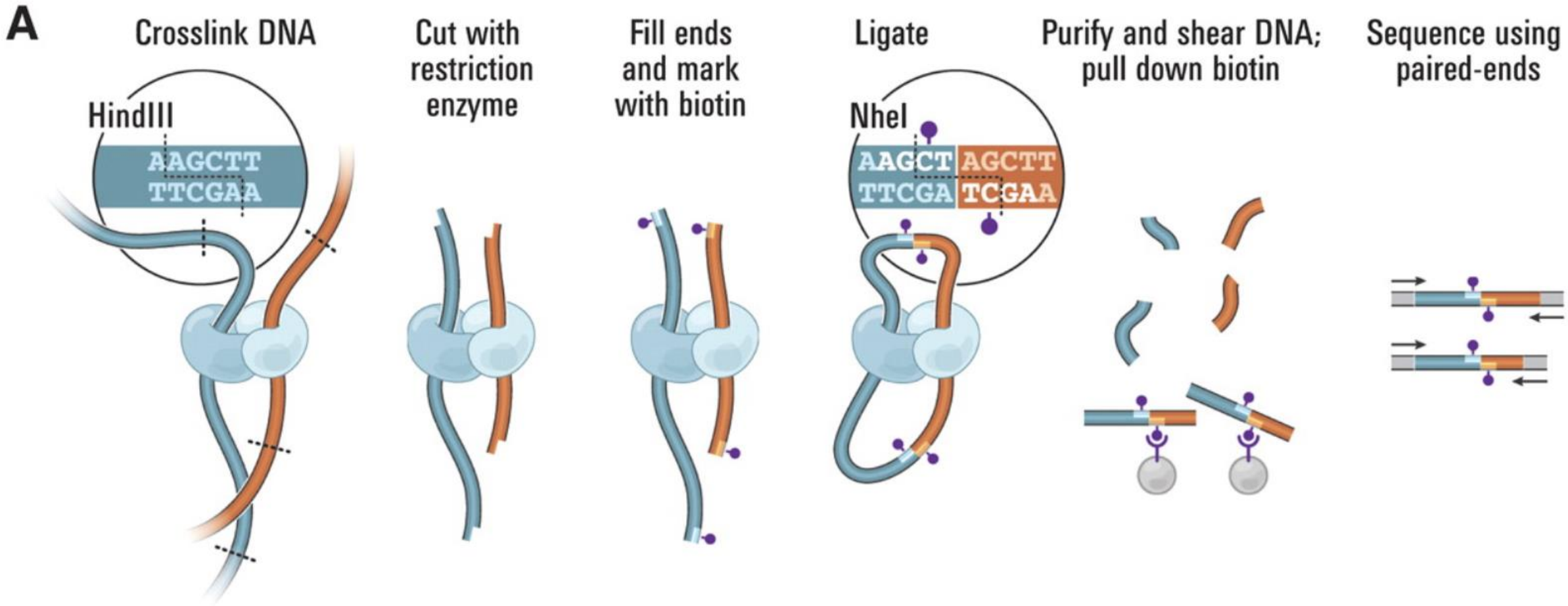
Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

EREZ LIEBERMAN-AIDEN, NYNKE L. VAN BERKUM, LOUISE WILLIAMS, MAXIM IMAKAEV, TOBIAS RAGOCZY, AGNES TELLING, IDO AMIT, BRYAN R. LAJOIE, PETER J. SABO, MICHAEL O. DORSCHNER, RICHARD SANDSTROM, BRADLEY BERNSTEIN, M. A. BENDER, MARK GROUDINE, ANDREAS GNIRKE, JOHN STAMATOYANNOPOULOS, LEONID A. MIRNY, ERIC S. LANDER, AND JOB DEKKER fewer

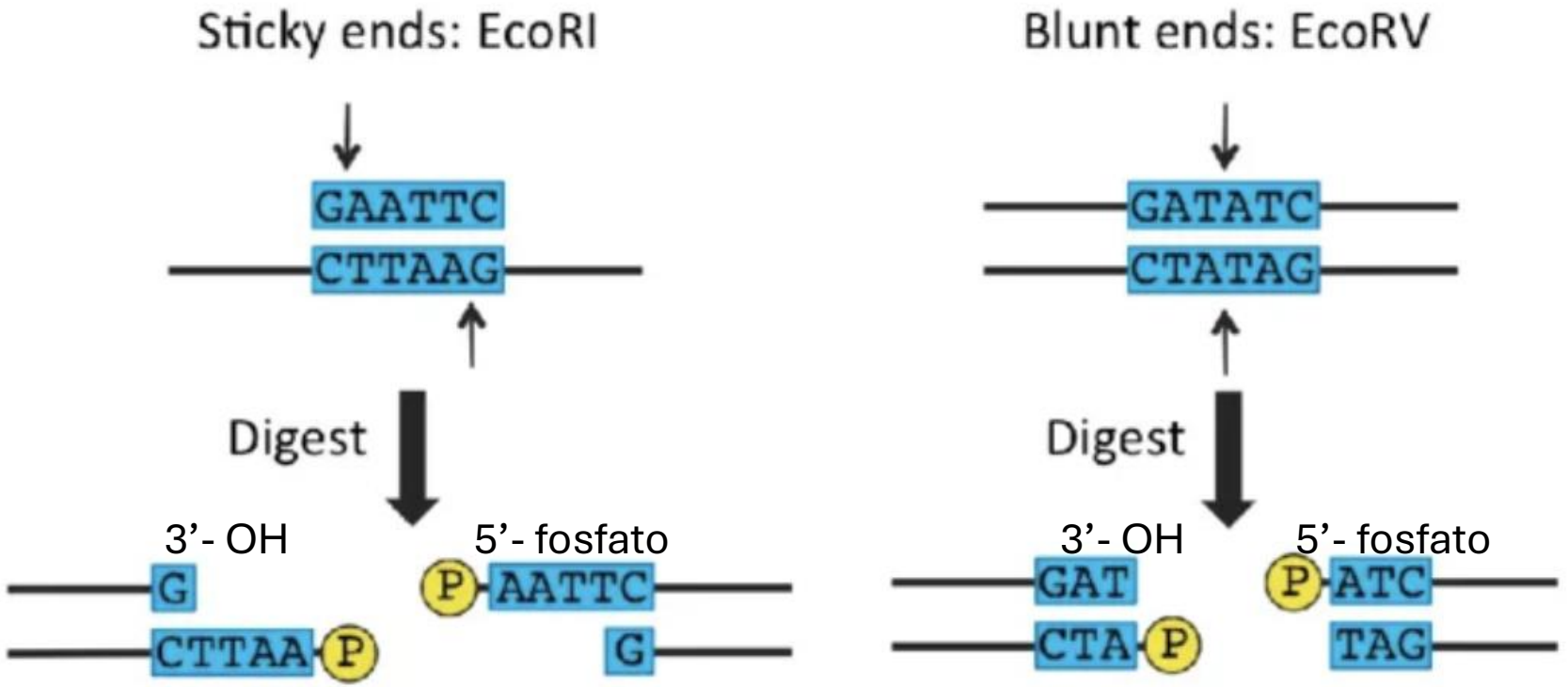
Authors Info & Affiliations

SCIENCE • 9 Oct 2009 • Vol 326, Issue 5950 • pp. 289-293 • DOI: 10.1126/science.1181369

O experiment de Hi-C

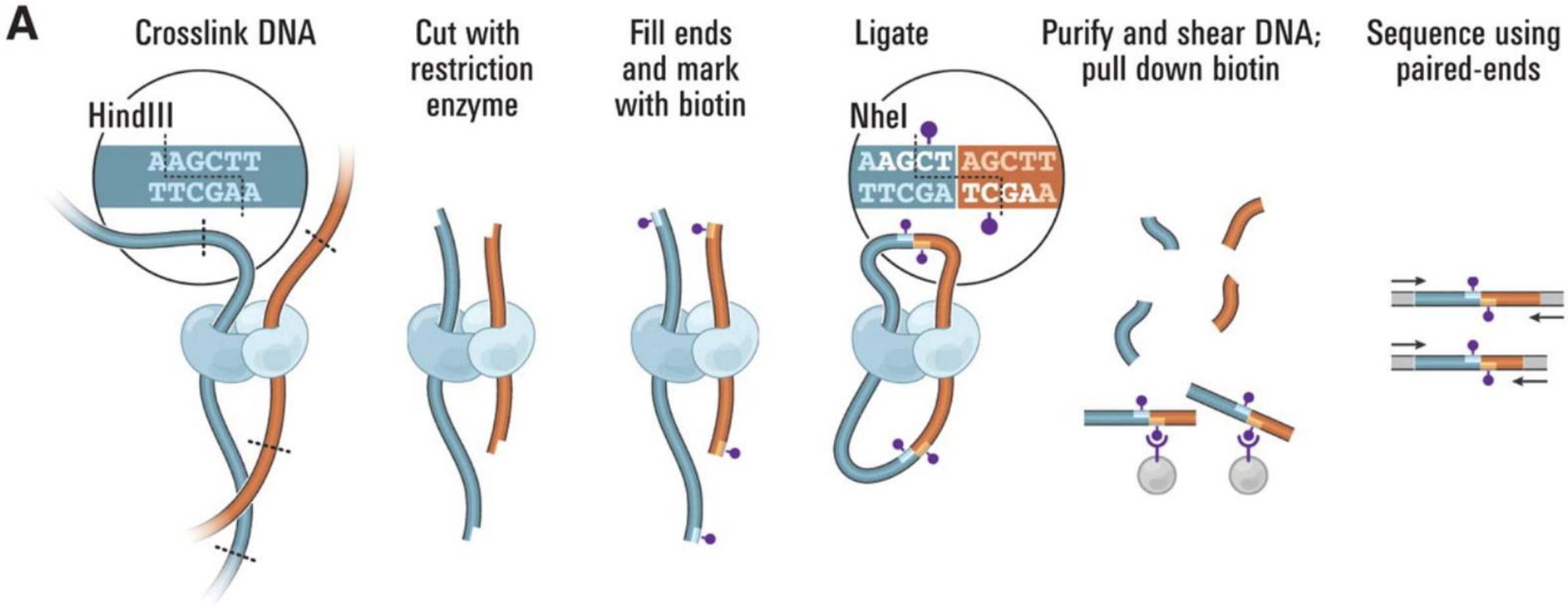


Preenchimento com DNA polimerase

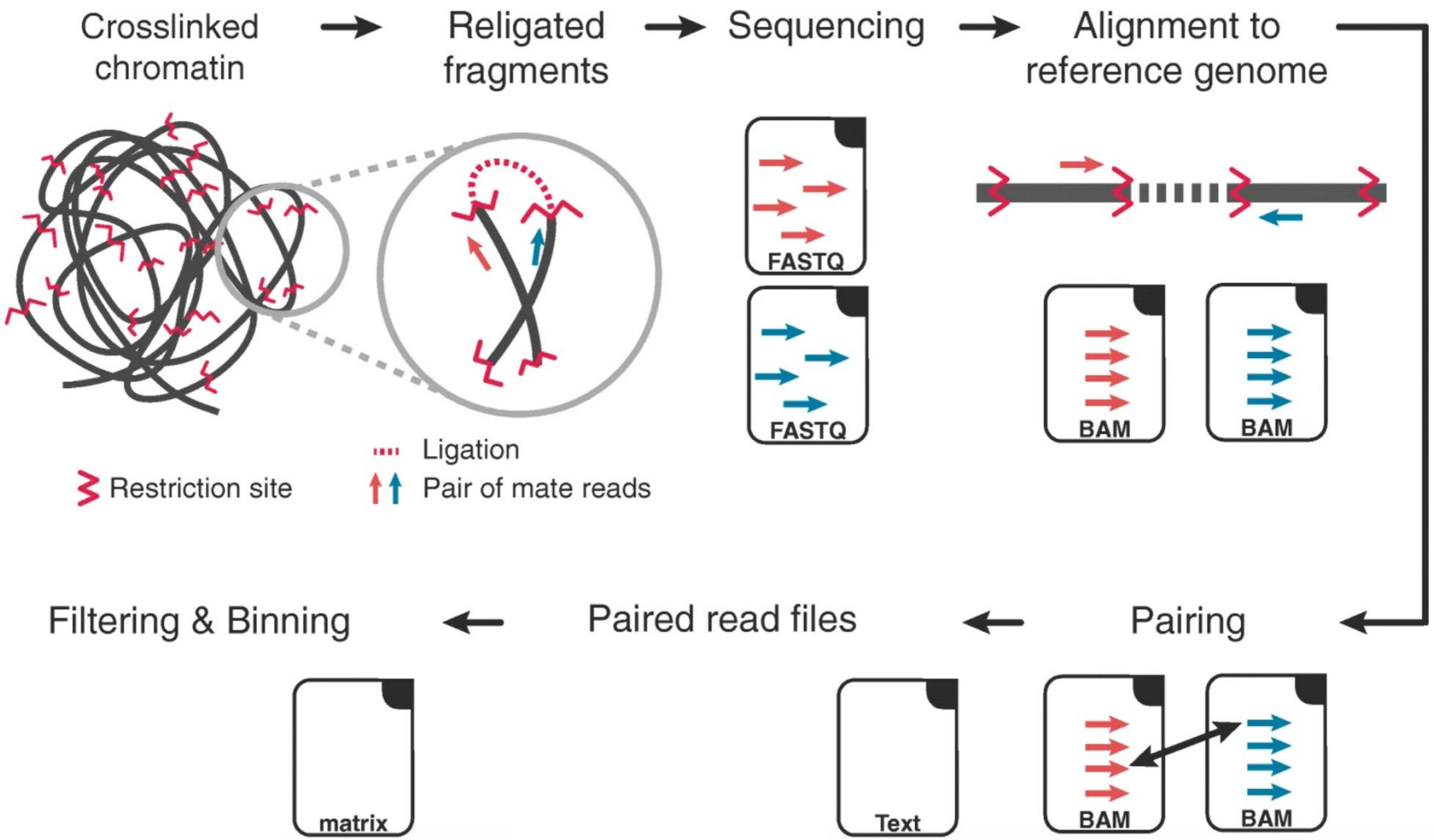


Bertero et al, 2017

O experiment de Hi-C

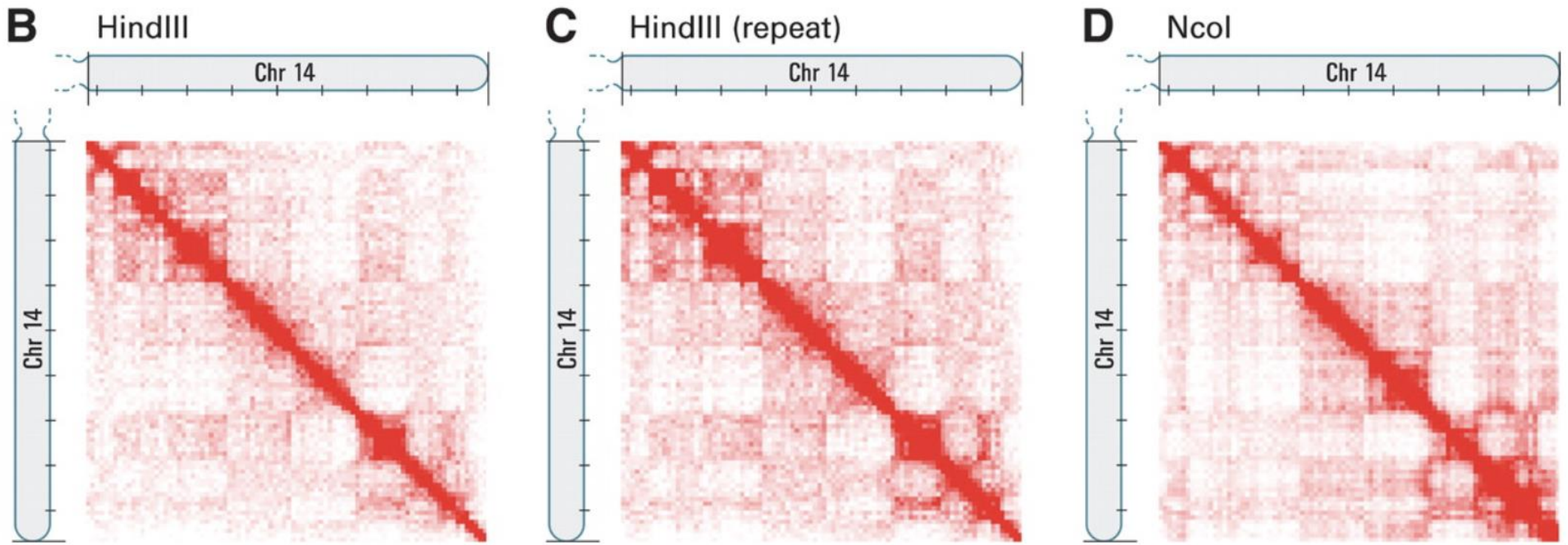


Sequenciamento e mapeamento dos pares de sequências num genoma de referência



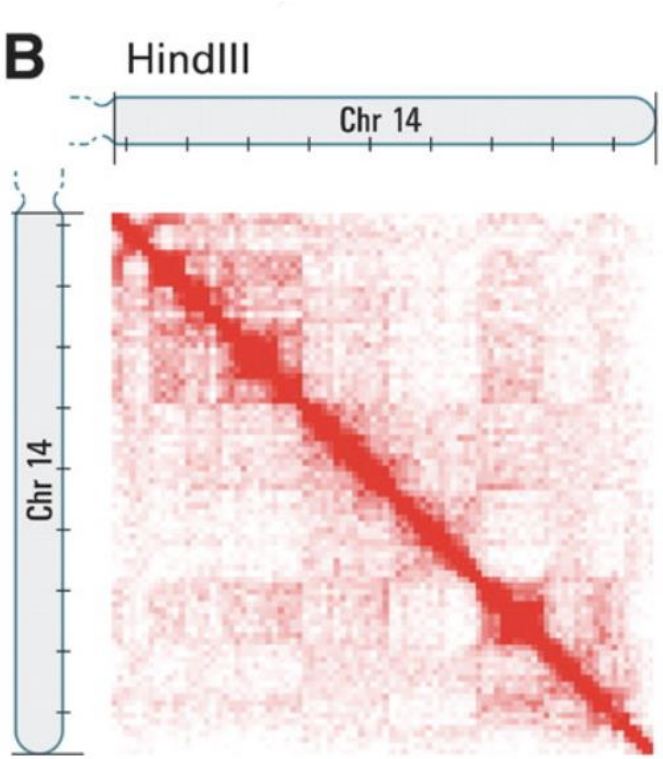
Pal et al, 2018

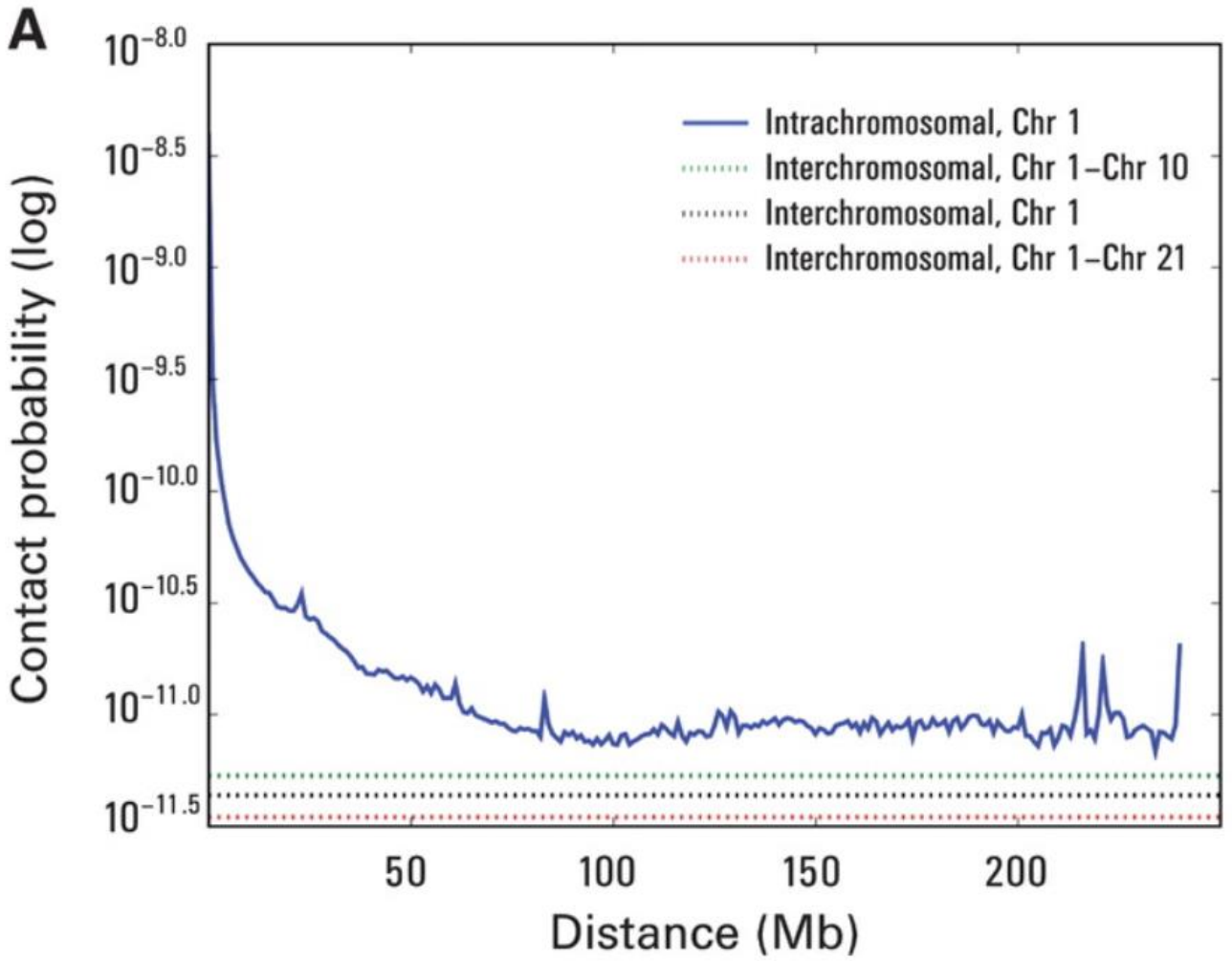
Mapas de contato a depender da enzima de digestão



Frequência ou probabilidade de contatos

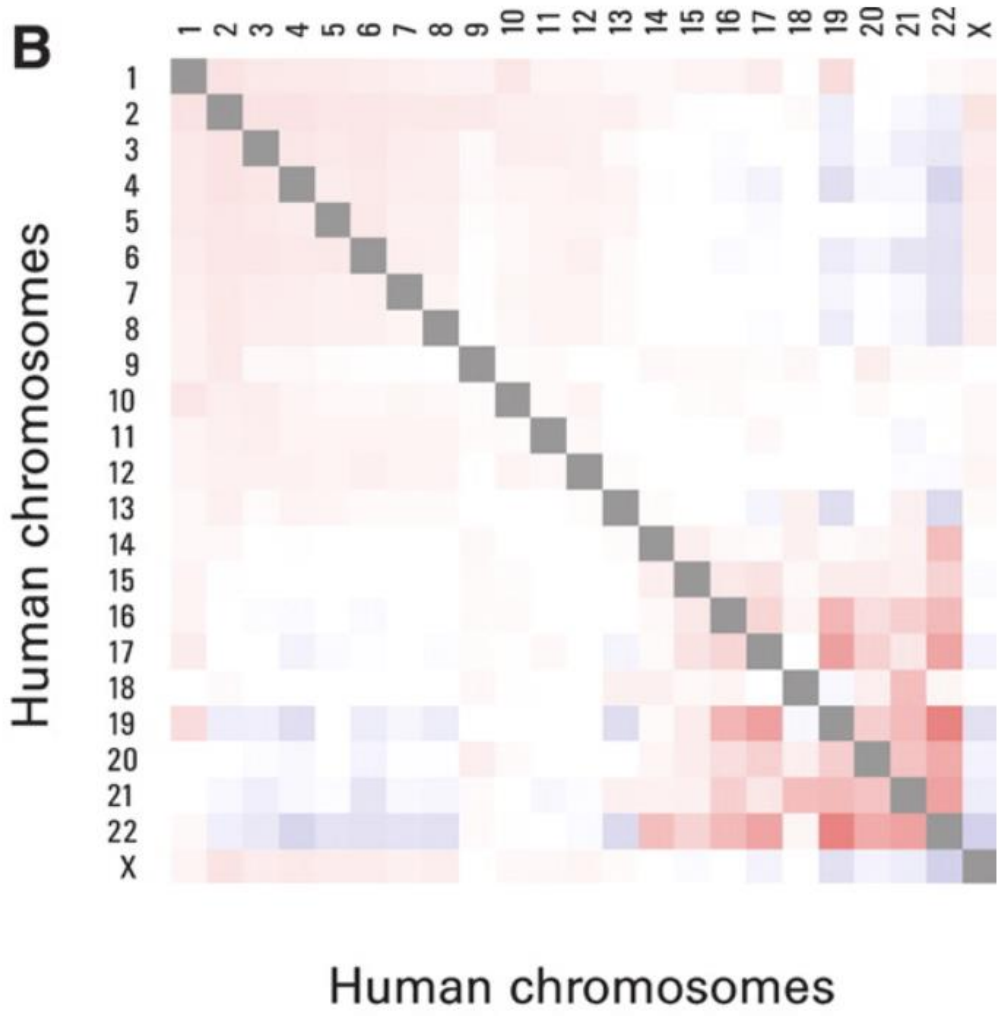
Média de contatos por distância entre sítios ligados





Padrão global, geral de contatos

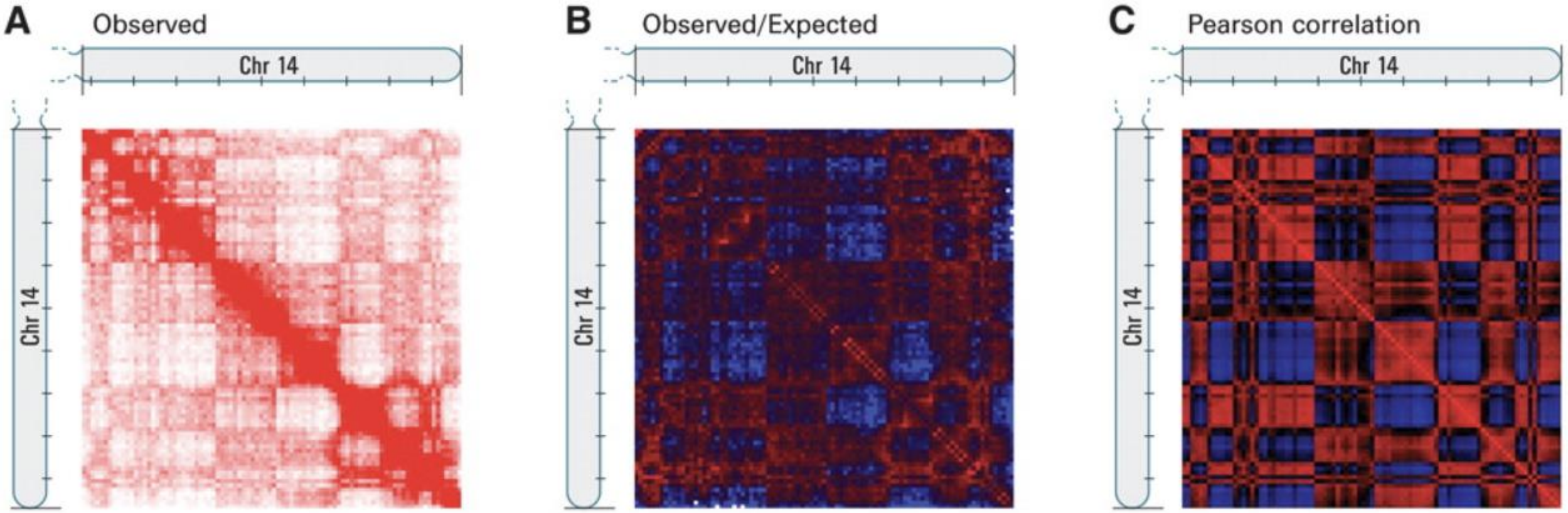
Frequência ou probabilidade de contatos



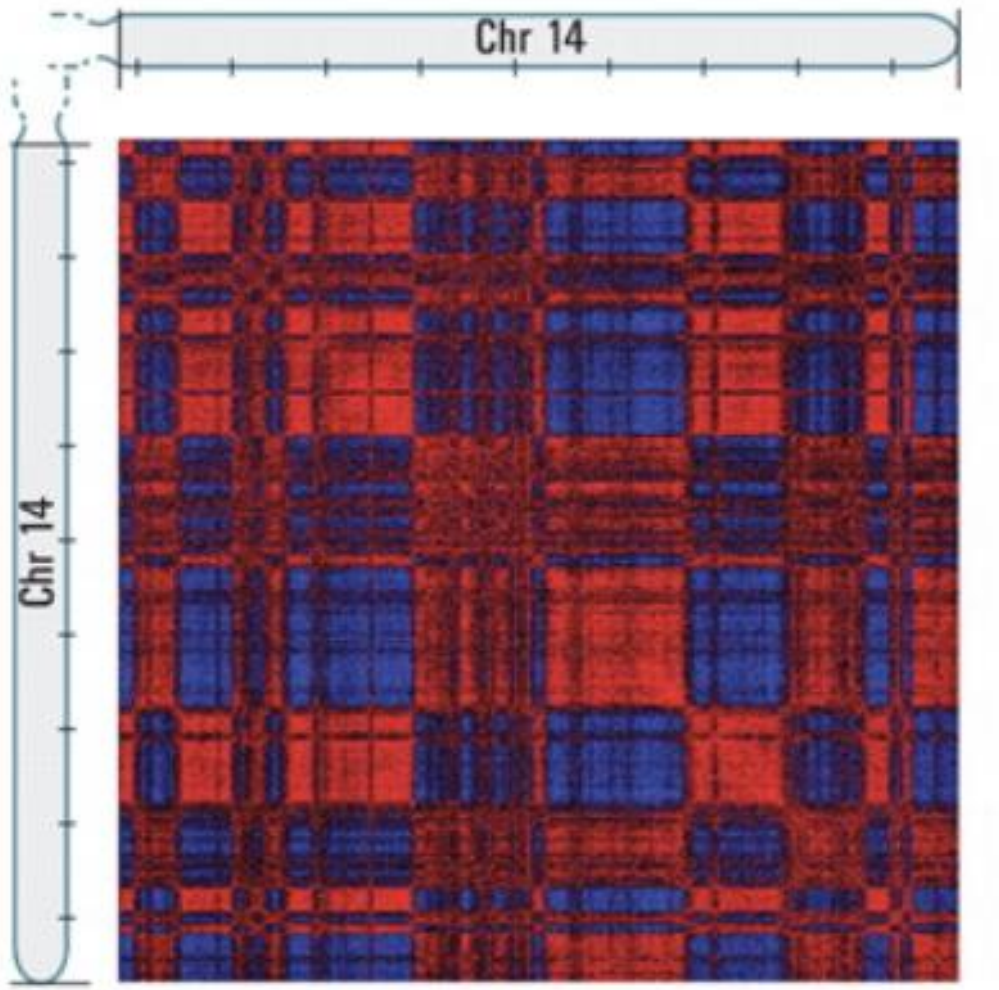
$$\frac{Contatos\ observados}{Contatos\ esperados\ (média\ de\ contatos\ por\ distância)}$$

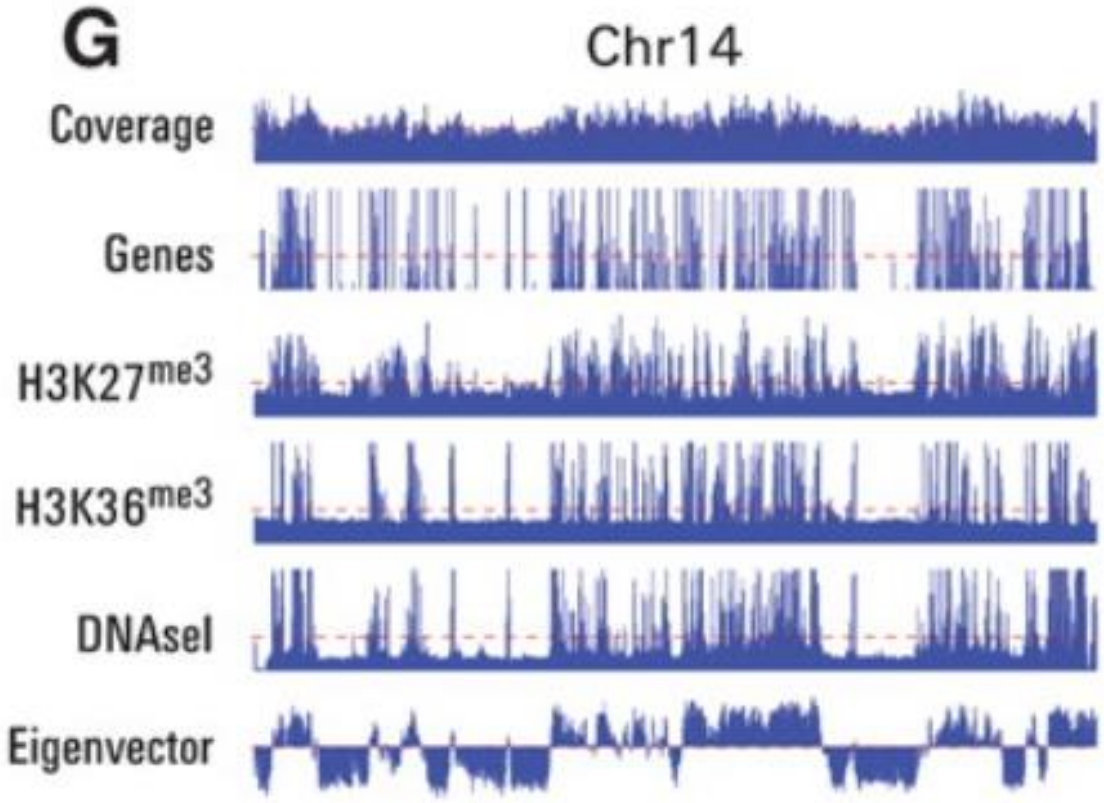
Menos que o esperado – azul
Mais que o esperado - vermelho

Enfatizando diferenças locais

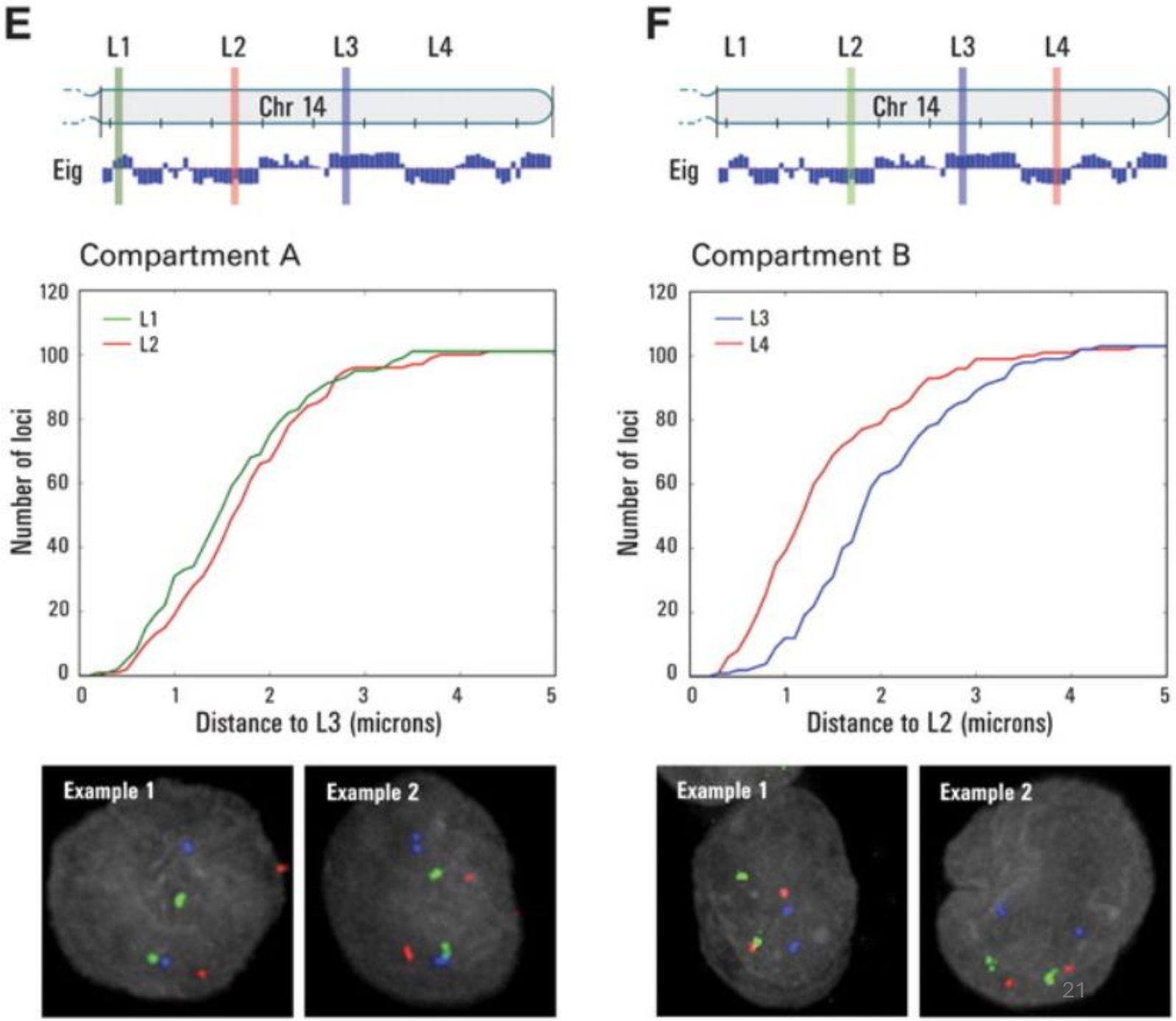


Compartimentos



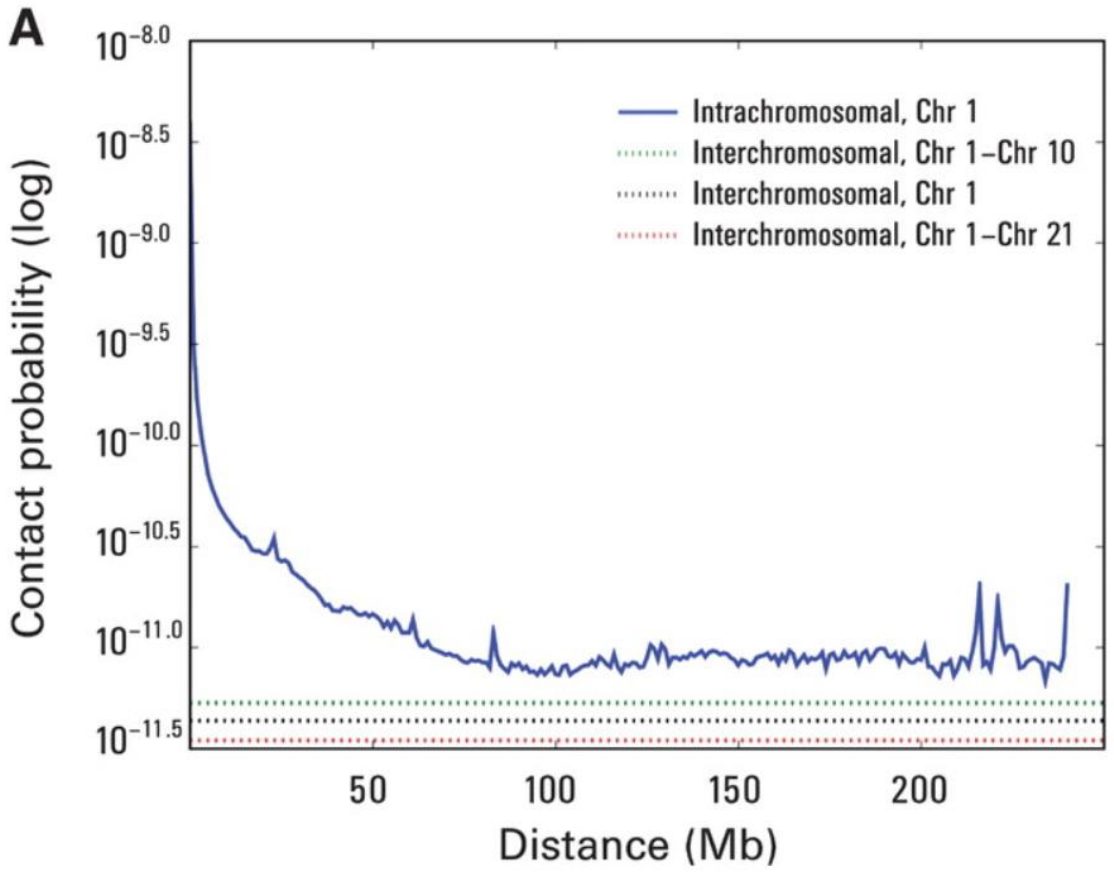


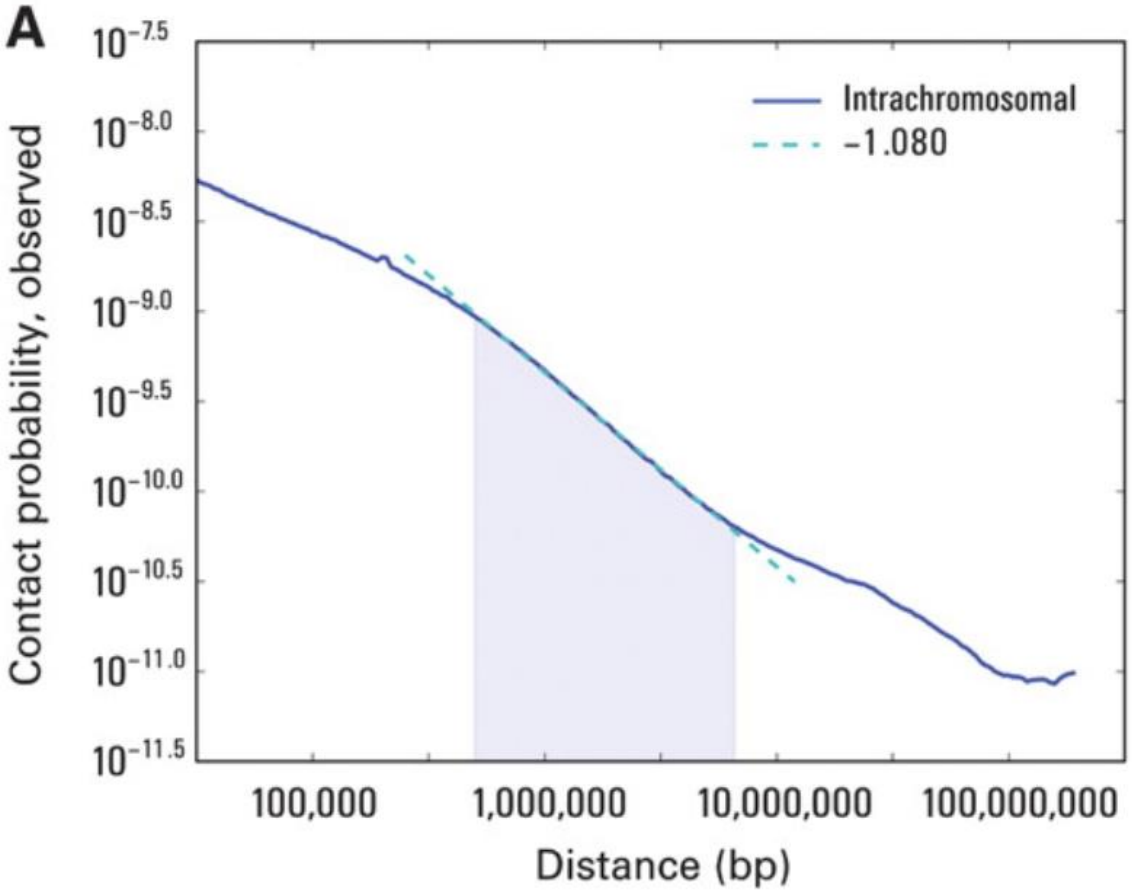
FISH de sítios específicos



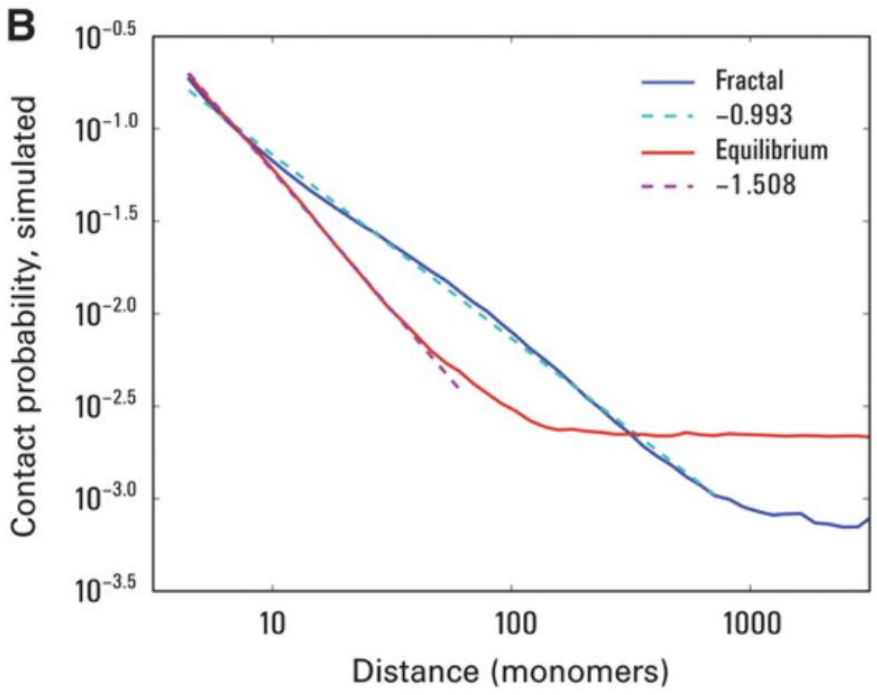
Frequência de contatos – padrão global

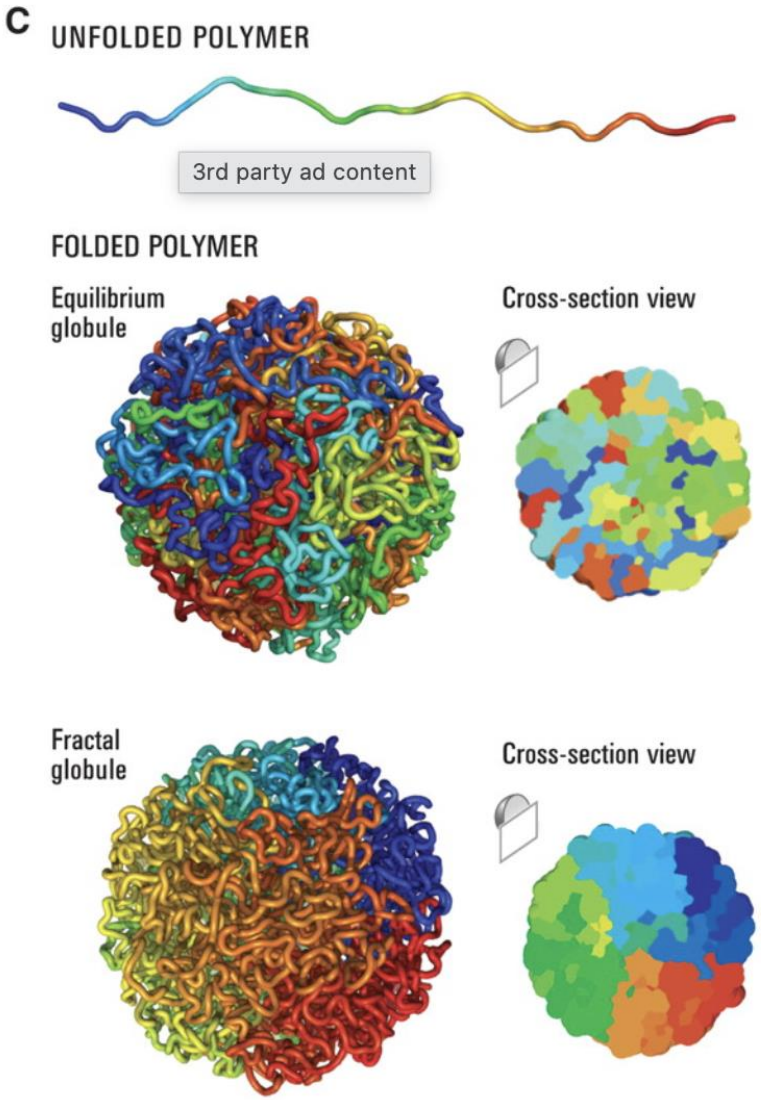
São a mesma coisa em escalas diferentes

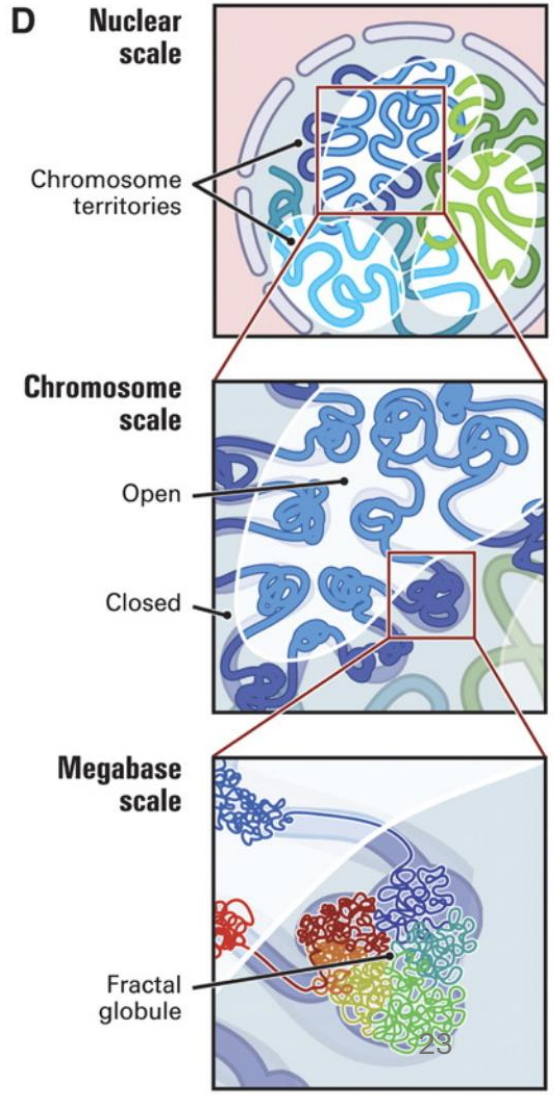




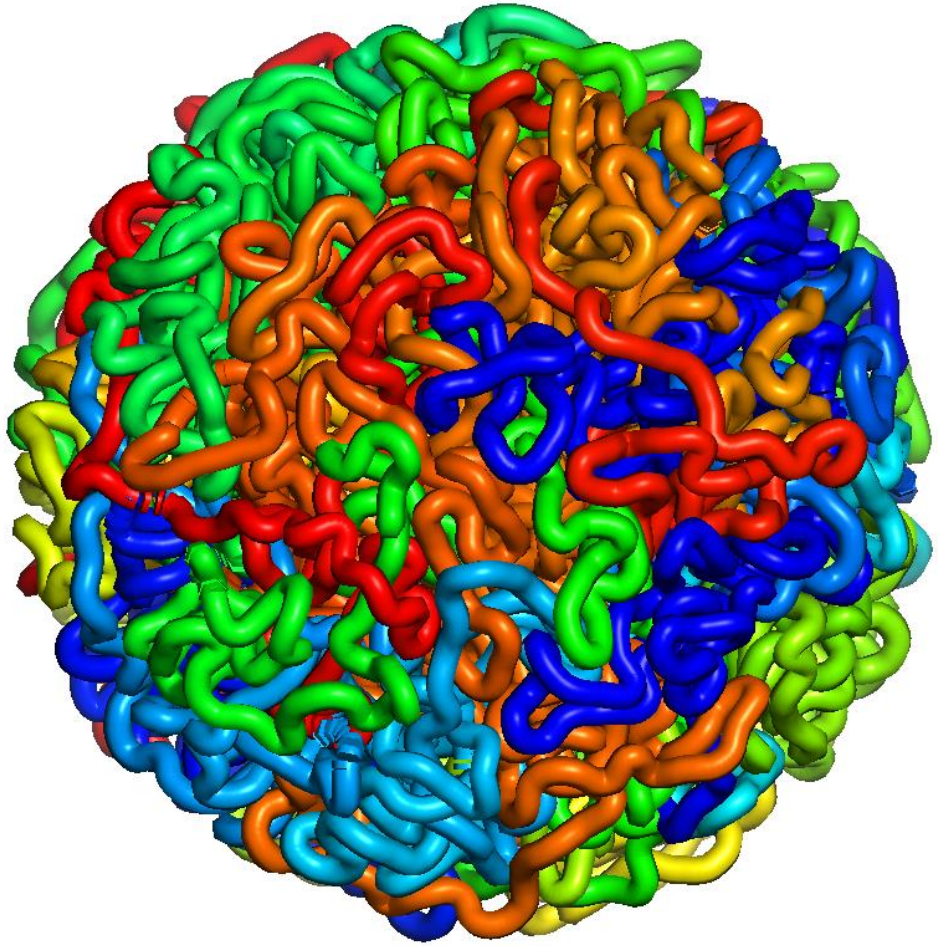
O arranjo da cromatina não é aleatório!

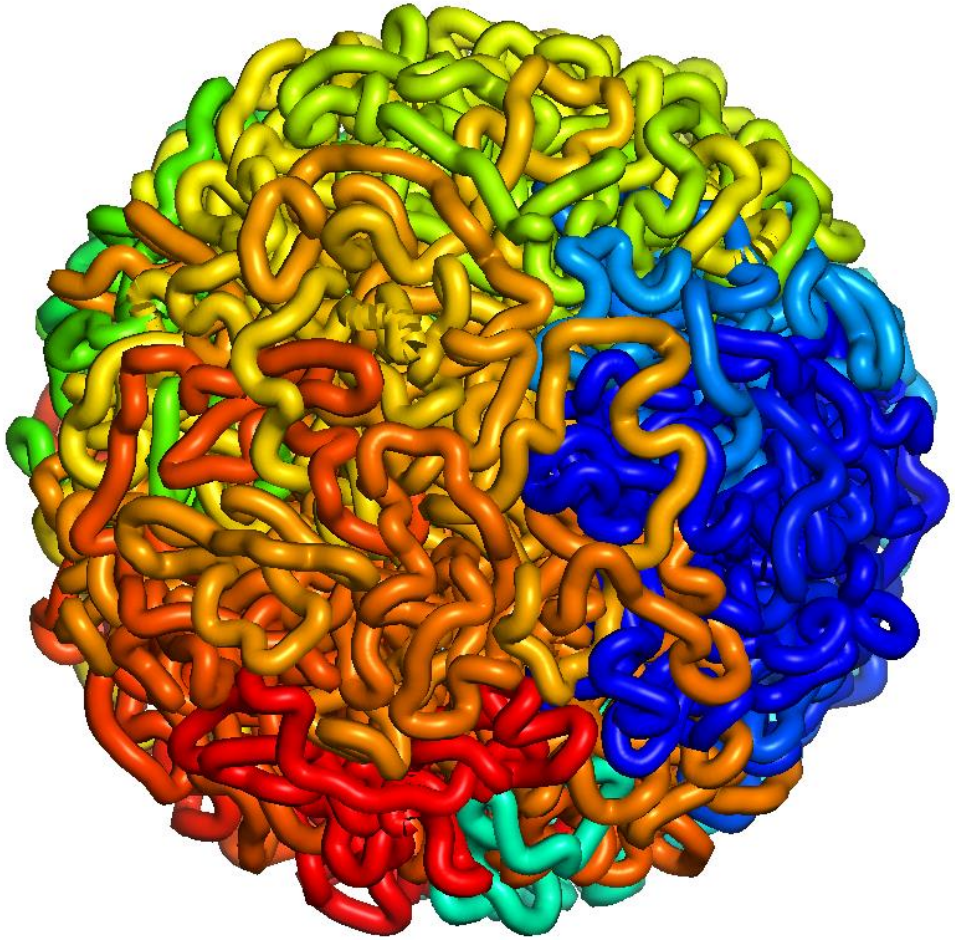






Equilibrium e fractal





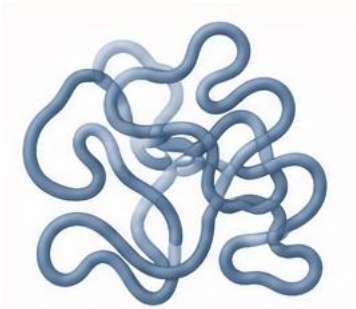
No pymol:

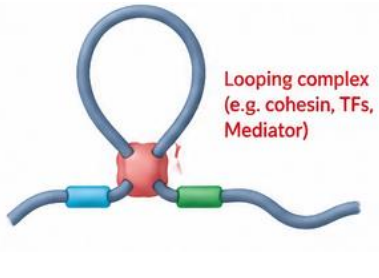
equilibrium.cif

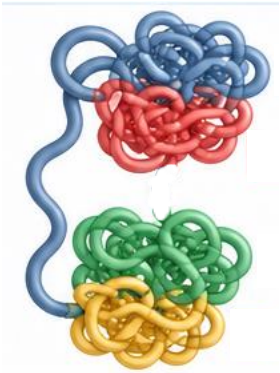
fractal.cif

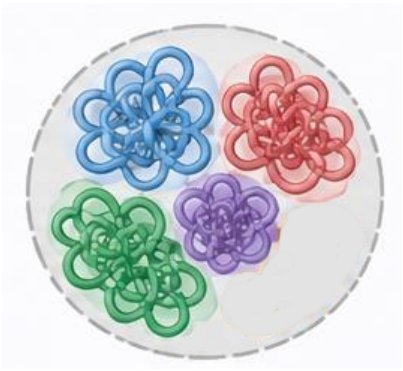
Exercício – combinar a conformação e o mapa de contatos

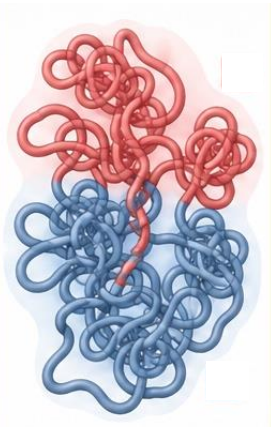
*Feito com IA



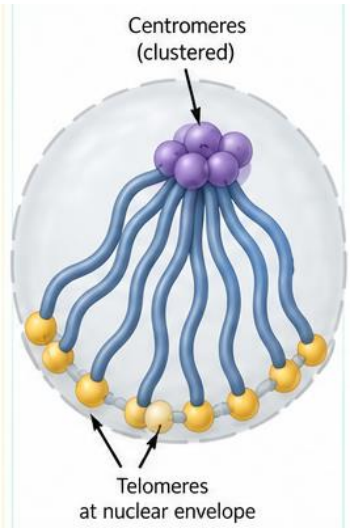


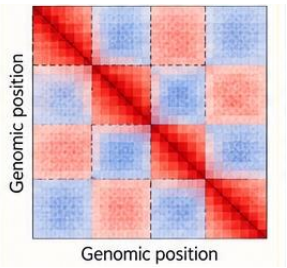


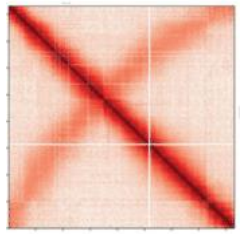


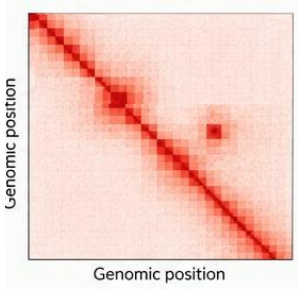


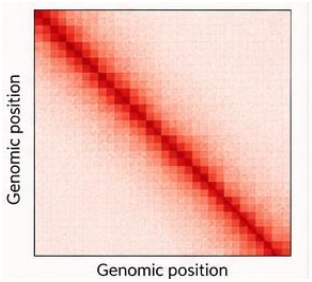


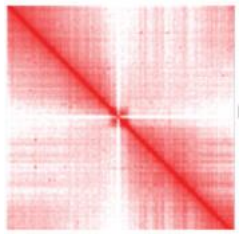


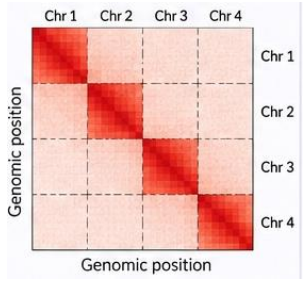


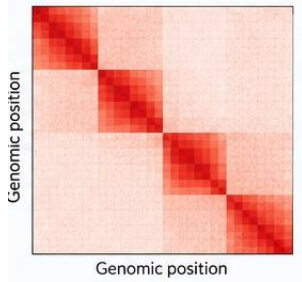




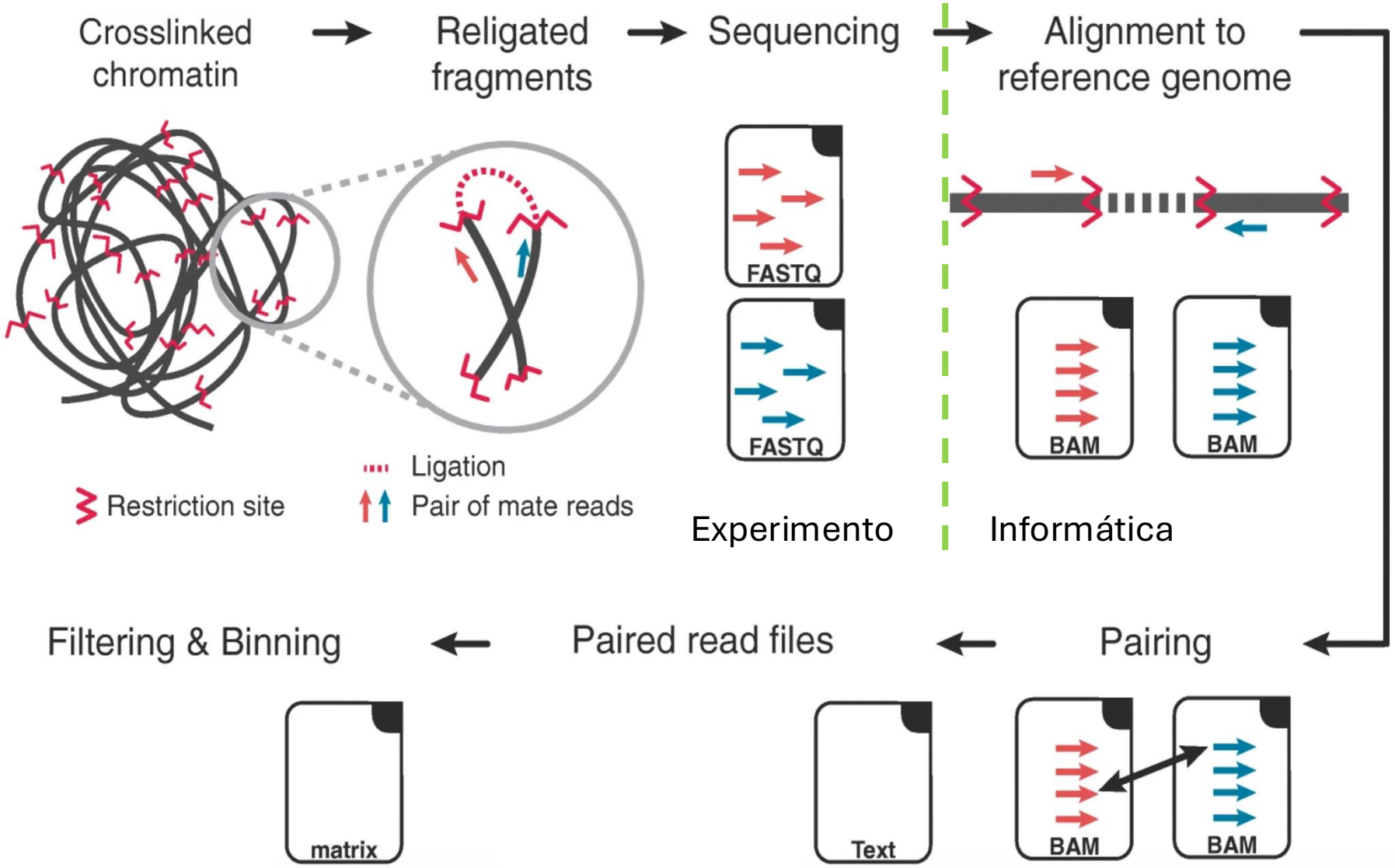








Como construir o mapa de contatos



Pal et al, 2018